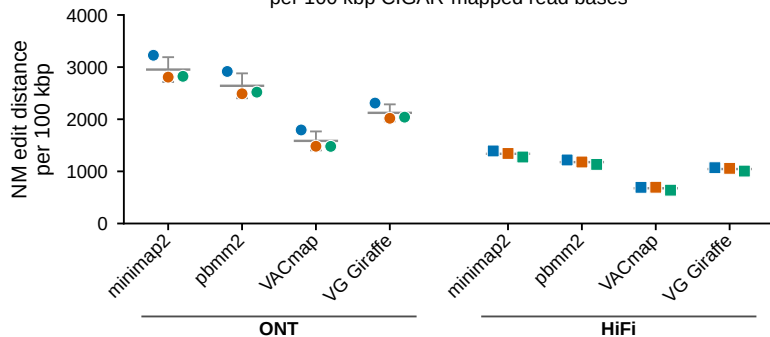


Alignment-based

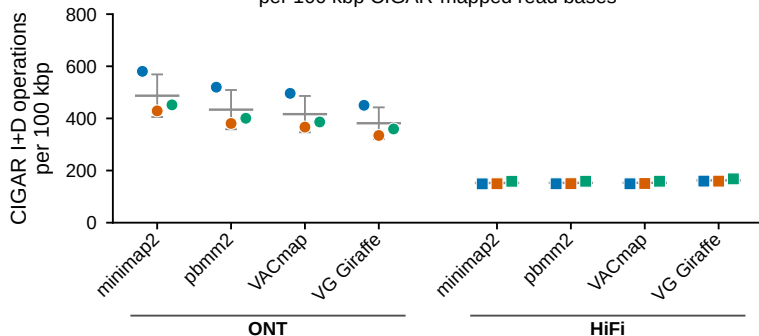
a Base-level disagreement

Reads: NM edit distance (substitutions + inserted + deleted bases) per 100 kbp CIGAR-mapped read bases



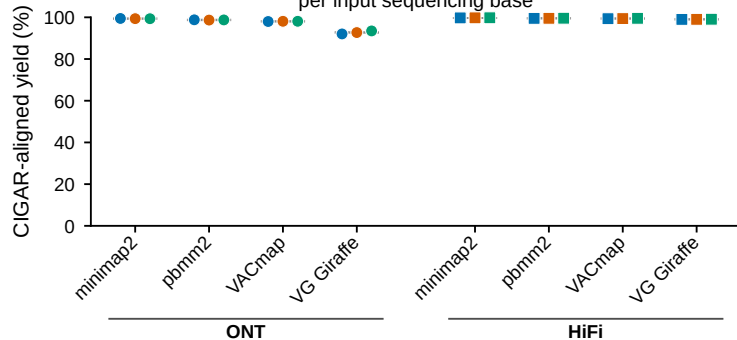
b Indel disagreement

Reads: CIGAR I + D operations (events, 1–300 bp; not bases) per 100 kbp CIGAR-mapped read bases



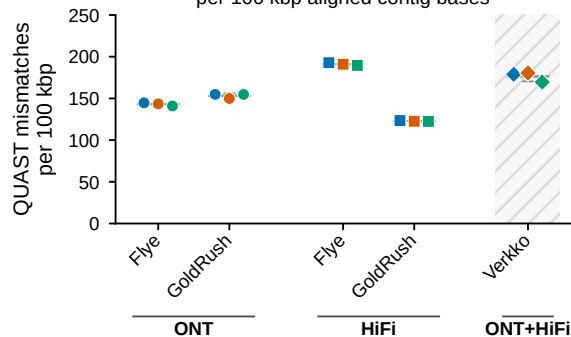
c Alignment yield and assembly completeness

Reads: CIGAR-mapped read bases per input sequencing base

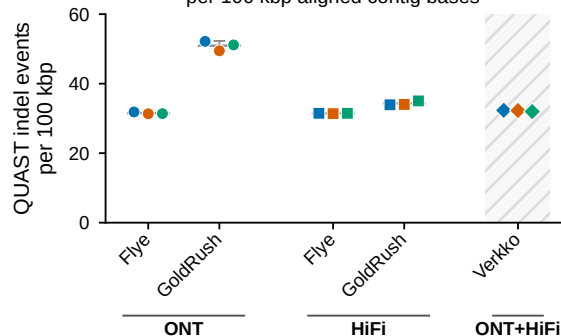


Assembly-based

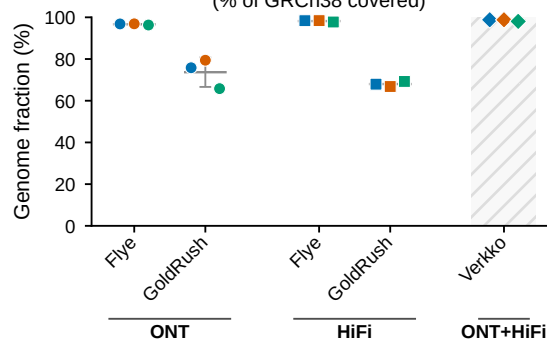
Contigs: QUASt mismatches (substitutions only) per 100 kbp aligned contig bases



Contigs: QUASt indel events per 100 kbp aligned contig bases



Contigs: QUASt genome fraction (% of GRCh38 covered)



● HG002 ● HG004 ■ HiFi — Mean ± SD across 3 samples
 ● HG003 ● ONT ◆ ONT+HiFi ▨ Verkko: both haplotypes in one FASTA